

Anshaditya Sharma

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Education

Vellore Institute of Technology (VIT)

Chennai, Tamil Nadu

B.Tech in Computer Science and Engineering; CGPA: 8.15/10

Aug 2023 – May 2027

- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems, Operating Systems, Computer Networks, Machine Learning, Computer Architecture, Applied Cryptography

Technical Skills

Languages: Java, Python, C++, C, JavaScript, TypeScript, Go, SQL, HTML/CSS

Frameworks & Libraries: React.js, Node.js, Express.js, FastAPI, TensorFlow, NLTK, NumPy

Systems & Infrastructure: MongoDB, Firebase/Firestore, Redis, Docker, GitHub Actions (CI/CD), REST API Design, WebSockets

Core CS & Security: Data Structures & Algorithms, System Design Fundamentals, Genetic Algorithms & Evolutionary Computation, Compiler Design (LALR/CLR/SLR Parsing)

Embedded & Hardware: ESP32, Arduino, Embedded C/C++, Sensor Integration, Servo/Motor Control

Tools & Practices: Git, Postman, Swagger/OpenAPI, JWT Auth, Linux, Unit Testing

Projects

Semi-Autonomous VLM-Controlled Rover | *ESP32-CAM, Arduino C++, Google Gemini Vision API – Final Year Project*

- Designed and built a rover on an ESP32-CAM with a pan-tilt camera mount and N20 gear motors, controllable manually or via natural-language commands over Wi-Fi.
- Integrated Google's Gemini Vision-Language Model to interpret live camera frames and generate real-time driving decisions for obstacle avoidance and target navigation, replacing hardcoded computer-vision rules with learned scene understanding.
- Authored a formal research report, "*Vision-Language Model Integration for Semi-Autonomous Mobile Robot Control*," documenting the system architecture, control loop design, and evaluation methodology.

EvoCreatures – Evolving Virtual Creatures via Genetic Algorithms | *Python, pymunk, NumPy, Three.js* demo | code

- Built a genetic algorithm that evolves locomotion controllers for an articulated creature in a 2D rigid-body physics simulation (pymunk), producing a coordinated crawling gait entirely through selection and mutation with zero scripted motion.
- Designed an exploit-resistant fitness function penalizing energy use and physically implausible "launch" gaits, combined with tournament selection, elitism, uniform crossover, and annealed Gaussian mutation with random immigrants to maintain genetic diversity.
- Improved best-genome fitness from 0 to 13.9 and crawl distance from ~7m to 21m over 30 generations in a fully deterministic, seeded run; shipped a live Three.js replay viewer with a real-time evolution chart, deployed statically via GitHub Pages.

LectureFind – AI-Powered YouTube Lecture Navigator | *React, FastAPI, Python, NLTK, Chrome Extension* link

- Engineered a semantic search engine that pinpoints concept-level timestamps inside YouTube lecture transcripts by combining WordNet synonym expansion with Wu-Palmer similarity scoring, eliminating manual scrubbing through hour-long videos.
- Built a temporal-clustering algorithm that merges matches within a 45-second window into coherent time ranges and surfaces the top 5 semantically related topics when zero direct matches exist.
- Shipped a FastAPI backend and Manifest V3 Chrome Extension that jumps the live YouTube player to the matched timestamp in a single click.

Automated Trading Order Execution CLI | *Python, Binance Futures API, unittest* link

- Built a CLI trading tool that places and validates MARKET, LIMIT, and STOP orders on Binance Futures by cross-checking symbol, quantity, and price against live exchange step-size rules, preventing malformed order submissions.
- Decoupled the API client, order validators, and CLI layers, then verified correctness with an automated unittest suite covering both BUY and SELL order flows.
- Instrumented structured request/response logging for every order to support auditability across live and mock trading runs.

Achievements & Leadership

Solved 175+ Data Structures & Algorithms problems on LeetCode (Java, C++); achieved a 2-star rating on CodeChef.

Served as Cultural Lead for the Uttar Pradesh Cultural Club, coordinating cross-functional event management, logistics, and team operations beyond the cultural portfolio.

Won multiple prizes across national and university-level hackathons in real-time systems and hardware-software integration tracks, leading teams of 4–6 members end-to-end from ideation to live demo.

Conducted peer-learning sessions on Data Structures, Git workflows, and API design for junior students at VIT Chennai.

Certified in Kotlin; completing an intensive Full-Stack, Web3, and DevOps engineering cohort covering production deployment and CI/CD pipelines.